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Utility Fairness in Contextual Dynamic Pricing with Demand Learning

Xi Chen,[a,*] David Simchi-Levi,[b] Yining Wang[c]

[a] Leonard N. Stern School of Business, New York University, New York, New York 10012; [b] Institute for Data, Systems and Society, Operations Research Center, Department of Civil and Environmental Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139; [c] Naveen Jindal School of Management, University of Texas at Dallas, Richardson, Texas 75080

*Corresponding author

Contact: xc13@stern.nyu.edu, https://orcid.org/0000-0002-9049-9452 (XC); dslevi@mit.edu (DS-L); yining.wang@utdallas.edu, https://orcid.org/0000-0001-9410-0392 (YW)

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Abstract. This paper introduces a novel contextual bandit algorithm for personalized pricing under utility fairness constraints in scenarios with uncertain demand, achieving an optimal regret upper bound. Our approach, which incorporates dynamic pricing and demand learning, addresses the critical challenge of fairness in pricing strategies. We first delve into the static full-information setting to formulate an optimal pricing policy as a constrained optimization problem. Here, we propose an approximation algorithm for efficiently and approximately computing the ideal policy. We also use mathematical analysis and computational studies to characterize the structures of optimal contextual pricing policies subject to fairness constraints, deriving simplified policies that lay the foundations of more in-depth research and extensions. Further, we extend our study to dynamic pricing problems with demand learning, establishing a nonstandard regret lower bound that highlights the complexity added by fairness constraints. Our research offers a comprehensive analysis of the cost of fairness and its impact on the balance between utility and revenue maximization. This work represents a step toward integrating ethical considerations into algorithmic efficiency in data-driven dynamic pricing.

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Keywords: contextual bandit • demand learning • dynamic pricing • minimax regret • utility fairness

1. Introduction

Personalized dynamic pricing, a pivotal strategy in modern revenue management, leverages customer-specific data to tailor prices. This approach maximizes revenue by aligning prices with individual customer value perceptions and willingness to pay. The advent of big data analytics and machine learning has significantly advanced the capabilities of personalized dynamic pricing. Businesses can now analyze vast amounts of data in real time to understand customer behaviors, preferences, and price sensitivities. This advancement has led to more sophisticated and granular pricing strategies, where prices can be tailored not only to segments of customers but to an individual consumer's contextual profile. In practical scenarios, e-commerce platforms typically implement personalized pricing strategies through the use of discounts or promotions rather than directly varying the prices.

However, the efficiency of dynamic pricing comes with complexities and challenges, particularly concerning fairness and ethical considerations. As businesses increasingly leverage customer data to optimize pricing strategies, there is a growing concern about the fairness of these practices, both from a customer perspective and within the regulatory framework. Regulators have intensified scrutiny over potential discrimination from pricing algorithms. For instance, regulatory bodies like the U.S. Federal Trade Commission (FTC) and the UK's Financial Conduct Authority (FCA) mandate that retailers disclose their pricing algorithms (FCA 2018, Smith 2020). These regulators then meticulously examine the design, inputs, and outputs of such algorithms to verify compliance with applicable fairness regulations. Retailers, in turn, must proactively implement fairness-aware dynamic pricing strategies. This approach is crucial to mitigate customer dissatisfaction and reputational damage arising from discriminatory pricing practices while concurrently adhering to regulatory standards.

Indeed, fairness is not just a moral imperative but a business necessity. Unfair pricing practices can lead to customer dissatisfaction, erosion of trust, and

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potential backlash, both from consumers and regulators. In the long term, these factors can adversely affect a company's reputation and profitability. Fair pricing policies must ensure that customers with similar characteristics and behaviors are offered similar prices. Therefore, in this paper, we consider a utility fairness (see Definition 1) concept, which requires the implemented pricing policy to offer similar prices to customers whose utility values are similar, even though their feature vector could be very different. Our concept of utility fairness is encapsulated by a Lipschitz continuity constraint, which measures the disparity between pricing policy and utility for a given pair of contextual vectors. The Lipschitz constant, denoted as δ_0, is indicative of the fairness constraint's intensity. Specifically, a lower value of δ_0 signifies a more stringent fairness requirement. The advantage of utility fairness is that it will discriminate against a specific feature for the "absolute" fairness. Instead, it treats the entire customer feature vector as a whole for the purpose of fairness.

By integrates fairness into personalized/contextual pricing, the framework balances profitability with ethical pricing practice. We study both the static (a.k.a., full-information) and multiperiod dynamic pricing problems. The proposed policies serve as a significant step in marrying ethical considerations with revenue objective. The structure of the paper and main contributions are summarized as follows.

1. We first study the static full-information setting, where the demand function is fully known. We formulate optimal pricing policy as a constrained optimization problem (see Equations (3) and (4)). To find a near-optimal optimal policy with general demand function, we introduce an approximation algorithm based on discretization of the utility space. This methodology offers a pragmatic solution to approximate the ideal policy within a polynomial time frame (see Section 4.1).

2. In addition to the discretization and approximation algorithm, we further provide analytical results that characterize the structure of optimal fair pricing policies, shedding light on the impact of fairness constraints (see Section 4.2). Under an assumption of the demand function (see Equation (9)) that is satisfied by many popular models, we establish several key properties of the optimal contextual pricing policy as a function of the utility (see Theorem 1). In particular, we demonstrate that the optimal contextual pricing policy is *monotonically nondecreasing*, *differential almost everywhere*, with the values of the pricing policy determined by fairness unconstrained prices when the differential of the underlying policy satisfies certain conditions. Furthermore, when the Lipschitz parameter δ_0 is not too large, we show that the optimal policy possesses a *linear structure* in the sense that the policy's offered prices are linear in customers' baseline utility values. This result yields several interesting insights. First, with a linear demand function, it fully defines the structure of the optimal contextual pricing policy under fairness constraints (see Remark 2). Second, for a linear demand function, we derive the closed-form expression for the *cost of utility fairness*, defined as the ratio of optimal revenue under fairness-constrained conditions to that under no-constraint conditions (see Theorem 2). This explicitly expresses the intrinsic tradeoff between fairness and profit. For other types of demands, we provide numerical illustration of the *cost of utility fairness* based on our discretization algorithm. This analysis helps in understanding the tradeoffs involved in implementing fairness in pricing strategies. We also remark that the structural results established here are highly nontrivial because the formulation of the optimal pricing policy (subject to fairness constraints) is essentially an infinite-dimensional optimization problem. Our proof relies on an infinite sequence of finite-dimensional optimization problems to approximate the infinite-dimensional problem, together with Lagrangian multiplier analysis and continuity/compactness in function spaces.

3. We further study the fair contextual pricing with unknown demand function in a T-period learning-while-doing setting (see Section 4) We first explore the regret lower bound to enhance our comprehension of the fairness constraint's influence in dynamic contextual pricing (see Section 5.1). A rather surprising result is the problem becomes intrinsically more difficult due to the fairness constraint. We establish the $\Omega(T^{2/3})$ regret lower bound, which is different from the standard $O(\sqrt{T})$ or $O(\log T)$ bounds in existing dynamic pricing or contextual bandit problems (Rusmevichientong and Tsitsiklis 2010, Li and Huh 2011). To understand the rationale behind this phenomenon, it is insightful to explore the structure of the optimal pricing policy in complete information setting (see Theorem 1). The structure reveals that the optimal price for a user with a specific context x is linearly in x, indicating a perfect colinearity between context vectors and offered prices in an optimal strategy. This presents a dilemma: Implementing a near optimal pricing policy results in high-colinearity between the context vector x and the price p, hindering actuate estimations of the parameters. Conversely, adopting pricing strategies to avoid such colinearity leads to suboptimal outcomes, deviating from the structure delineated in Theorem 1. This challenge results in an unconventional regret lower bound of $\Omega(T^{2/3})$, which underscores the impact of the fairness constraint from an information-theoretical standpoint.

4. We introduce a new contextual bandit algorithm designed for personalized pricing under utility fairness constraints and uncertain demand, featuring a rate-optimal regret upper bound of $O(T^{2/3})$ (see Section 5.2).

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This algorithm comprises two phases: The initial phase involves price experimentation, during which maximum likelihood estimates are formulated from the gathered data. The subsequent phase applies the upper-confidence bound (UCB) algorithm, where each arm represents a discretized price for the zero utility value, with prices for customers with other utility values set according to the optimal linear structure of fairness constrained policies (see Algorithm 1 for more details).

In summary, this paper provides a rigorous approach to integrate utility fairness into personalized contextual pricing. For both static and dynamic settings, we fully explore the cost of the fairness constraint in the optimal revenue and regret bound. We believe this work offers significant steps toward bridging the gap between algorithmic efficiency and ethical considerations in data-driven dynamic pricing. Our framework could potentially be applied and extended to other types of revenue management problems.

2. Related Works

This section briefly reviews two pertinent research areas: dynamic pricing and some recent development on fairness

2.1. Dynamic Pricing

With the growth of online retailing, dynamic pricing has emerged as a prominent research topic in recent years. For extensive surveys, see Bitran and Caldentey (2003), Elmaghraby and Keskinocak (2003), and Den Boer (2015). Our focus is on the single-product pricing problem. The foundational work of Gallego and Van Ryzin (1994) established the basis of dynamic pricing. Traditional models often assumed known demand information, but modern industries like fast fashion face unpredictable demand, sparking research on dynamic pricing with demand learning (Araman and Caldentey 2009; Besbes and Zeevi 2009; Farias and Van Roy 2010; Broder and Rusmevichientong 2012; Harrison et al. 2012; den Boer and Zwart 2013; Keskin and Zeevi 2014; Lei et al. 2014; Wang et al. 2014, 2021b; Chen et al. 2015, 2022c; Bastani et al. 2021; Miao et al. 2022; Chen and Wang 2023).

The surge in e-commerce has granted retailers unparalleled opportunities to analyze and cater to individual customer preferences. Thus, the personalized/contextual dynamic pricing has been adopted or considered in several key industries, including air travel, hotel reservations, insurance, and ride-sharing. The burgeoning practice within the industry has sparked a wealth of insightful research endeavors (Aydin and Ziya 2009; Lobel et al. 2018; Javanmard and Nazerzadeh 2019; Ban and Keskin 2021; Chen and Gallego 2021; Chen et al. 2022b, d, 2023a, 2024a; Wang et al. 2023; Fan et al. 2024; Luo et al. 2024).

Our research distinguishes itself by integrating fairness constraints into the realm of personalized dynamic pricing, uncovering that the optimal regret takes the rate of $O(T^{2/3})$, instead of the standard rate $O(T^{1/2})$ (Rusmevichientong and Tsitsiklis 2010, Li and Huh 2011). Our lower bound theorem validates the minimax regret, thereby underscoring the intrinsic challenges posed by the incorporation of fairness considerations into dynamic pricing models.

2.2. Fairness in Pricing

Fairness, extensively studied in economics, has recently gained significant attention in operations management. Fairness has been incorporated into operations problems, such as service priority, ads market resource allocation, online resource allocation, and dynamic rationing (Bateni et al. 2016, Chen and Wang 2018, Balseiro et al. 2021, Chen et al. 2022a, Manshadi et al. 2023). In pricing literature, fairness notion has been incorporated into game theoretical models for monopoly and duopoly markets (Li and Jain 2016, Kallus and Zhou 2021, Cohen et al. 2022).

Because of space constraints, an exhaustive review of fairness in operations management is beyond our scope. Instead, our discussion will center on a select few recent studies in fairness-aware dynamic pricing, noting that these studies have yet to integrate contextual or personalized information. The works by Cohen et al. (2022, 2021) address group fairness by segmenting customers into distinct groups, ensuring that the price gap between groups i and j at any time t (represented as $p_{i,t}$ and $p_{j,t}$, respectively) does not exceed a predefined threshold $\delta_{i,j}$ (i.e., $|p_{i,t} - p_{j,t}| \le \delta_{i,j}$). In this work, we significantly expand on the existing works by modeling users' groups *continuously*, with both user contexts and their utility values being continuous real numbers and the pairwise group fairness notion being superseded with Lipschitz continuity constraints that apply to continuous values. Moving beyond a static $\delta_{i,j}$, Chen et al. (2021) introduced a relative-gap fairness constraint and delves into nonparametric demand functions. Xu et al. (2023) considered the dynamic pricing under two types of fairness constraint simultaneously, a procedural fairness and a substantive fairness , and established the $O(\sqrt{T})$ regret bound. Further, Chen et al. (2024b) investigated dynamic network revenue management, employing the regularized fairness concept from Balseiro et al. (2021) to manage resource consumption during dynamic pricing. However, none of these studies on dynamic pricing with demand learning incorporate customer contextual information, which is the focal point of our paper.

2.3. Super-$\sqrt{T}$ Lower Bound

One of the important contributions made in this paper is to show that, when subject to utility fairness

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constraints, no learning-while-doing (bandit) algorithm could achieve an asymptotic regret significantly smaller than $O(T^{2/3})$, which contrasts most bandit and pricing settings in which $\widetilde{O}(\sqrt{T})$ or better regret is expected. In the literature, there are several other bandit settings that have super-$\sqrt{T}$ lower bounds. Here we do a brief survey of them and explain that why the reasons for such lower bounds in existing works are different from our settings.

When the underlying demand model or other mathematical objects needed to be learnt does *not* have simple parametric forms, and the overall problem lacks global convexity/concavity, an $\Omega(T^{3/5})$ or $\Omega(T^{3/4})$ regret lower bound might arise from the necessity of local smooth approximations. This is the case in the works of Wang et al. (2021a) and Chen et al. (2023b), and more generally in the continuum-armed (contextual) bandit literature (Bubeck et al. 2011, Gur et al. 2022, Hu et al. 2022). Because the demand model studied in this paper is a generalized linear model with a finite number of parameters and general regular conditions, the $\Omega(T^{2/3})$ regret bound established here is *not* from the lack of parametric forms.

When the application scenario limits the number of *action changes*, or places significant penalties on certain action switches (e.g., price protection, markdown pricing, or expensive capacity adjustments), a super-$\sqrt{T}$ regret lower bound might arise because of the bandit algorithm's inability to effectively explore certain arms. This is the case in the works of Jia et al. (2021), Feng et al. (2023), Chen et al. (2023c, 2020), and Simchi-Levi and Xu (2019), where the lower bounds range take the forms of $\Omega(T^{2/3})$ or a term polynomially higher than $\Omega(\sqrt{T})$, depending on the number of action changes/switches allowed. Although similar at the surface, such lower bounds are fundamentally different from our results because in our problem there is no limitation or penalty at all on how frequently the bandit algorithm could change its prices or pricing policies over time.

A regret lower bound arises in our setting due to the specific structure of the optimal fairness-constrained contextual pricing policy. This optimal policy is completely linear in the contexts. As a result, a bandit algorithm must make tradeoffs between exploration of suboptimal prices, or exploitation that inevitably introduces colinearity among covariates. Such a phenomenon is quite rare in bandit-related pricing literature, and our lower bound result and proof techniques could potentially see applications to other fairness-constrained bandit problems as well.

3. Models and Assumptions

We adopt a generalized linear model to model the probabilistic realizations of demands given customer contexts $x \in \mathcal{X} \subseteq \mathbb{R}^d$ and offered price $p \in [\underline{p}, \overline{p}] \in [0,1]$. More specifically, the realized demand $y \in [0,1]$ is being modeled by

$$\mathbb{E}[y|x,p] = f(x^\top \theta_0 - \alpha_0 p), \tag{1}$$

where $f(\cdot)$ is a known "link" function. Examples of popular link functions include the identity function $f(u) = u$ corresponding to the linear demand model, the logistic function $f(u) = e^u/(1+e^u)$ corresponding to the logistic demand model, the exponential function $f(u) = 1 - e^{-u}$ corresponding to the exponential demand model, and so on.

Let

$$r_u(p) = pf(u - \alpha_0 p) \tag{2}$$

be the expected revenue function for a customer with baseline utility value $u = x^\top \theta_0$. We impose the following technical assumptions throughout this paper.

Assumption 1 (Boundedness). *There exist constants $B \le \widetilde{B} < \infty$ such that for all $x, p \in \mathcal{X} \times [\underline{p}, \overline{p}]$, $x^\top \theta_0 \in [-B, B]$, $x^\top \theta_0 - \alpha_0 p \in [-\widetilde{B}, \widetilde{B}]$ and $f(x^\top \theta_0 - \alpha_0 p) \in [0,1]$.*

Assumption 2 (Smooth and Strongly Unimodal Revenue). *There exist constants $L_f < \infty$ and $\sigma_r > 0$ such that $|f(u)|$, $|f'(u)|, |f''(u)| \le L_f$ for all $u \in [-\widetilde{B}, \widetilde{B}]$. Furthermore, for every $u \in [-B, B]$, $r_u(\cdot)$ is unimodal,[1] and there exists a unique strictly interior revenue-maximizing price $p^*(u) = \arg\min_{p \in [\underline{p}, \overline{p}]} r_u(p) \in (\underline{p}, \overline{p})$ such that for every $p' \in [\underline{p}, \overline{p}]$, $r_u(p') \le r_u(p^*(u)) - \frac{\sigma_r}{2}(p' - p^*(u))^2$.*

Assumption 3 (Stochastic Contexts). *There exists an unknown underlying distribution $P_{\mathcal{X}}$ supported on $\mathcal{X}$, such that the context vector x of a customer is distributed independently and identically from $P_{\mathcal{X}}$ and that the baseline utility values $x^\top \theta_0$ are supported on a compact interval contained in $[-B, B]$. Furthermore, there exists a numerical constant $L_U < \infty$ such that for any measurable $U \subseteq [-B, B]$, $\Pr[x^\top \theta_0 \in U] \le L_U |U|/(2B)$, where $|U|$ is the Lesbesgue measure of $U \subseteq [-B, B]$.*

Assumption 1 assumes all model parameters are upper bounded by suitable constants, which is standard in the literature. Note that some of the upper bound parameters here, such as B or $\widetilde{B}$, may potentially scale polynomially with dimension d. Because we treat dimension as a constant in the low-dimensional setting studied in this paper, we shall not further elaborate on such potential dependency and instead deal with constants B and $\widetilde{B}$ throughout.

Assumption 2 assumes that the link function f is twice continuously differentiable with uniformly upper bounded derivatives. Because the expected revenue function admits the form in Equation (2), this together with Assumption 1 implies that $r_u(\cdot)$ is twice continuously differentiable with bounded derivatives too. Such smoothness assumption is standard in the literature. Furthermore, Assumption 2 assumes that, for every baseline utility value u, the expected

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revenue function $r_u(p)$ has a unique maximimzer $p^*(u)$ and decays quadratically as the price deviates from $p^*(u)$ on both directions. Such an assumption is implied by concavity in expected demand rates and monotonicity of link functions, which are satisfied by most popular demand functions (linear, logistic, exponential, etc.) (Li and Huh 2011). Notably, Assumption 2 does *not* imply that $r_u(\cdot)$ is concave in price, a condition that is usually violated for nonlinear link functions.

Assumption 3 assumes that users' context vectors $x \in \mathcal{X}$ are independently and identically distributed with respect to an underlying distribution $P_{\mathcal{X}}$ supported on $\mathcal{X}$. Although in general, contextual bandit the context vectors do not have to be stochastic and could even be adaptively adversarial, stochasticity of contexts is necessary here to define the optimal pricing policy subject to utility fairness constraints, because there is no longer pointwise optimal prices when the entire pricing policy is subject to fairness constraints. The second part of Assumption 3 further assumes that the marginal distribution of $x^\top\theta_0$ admits a well-defined probability density function, which is a mild condition and is satisfied provided that $P_{\mathcal{X}}$ does not have mass concentrated on a measure-zero subset of $\mathcal{X}$.

3.1. Fair Pricing Policies

In this section, we give the formal mathematical definition of (utility) fairness for a certain pricing policy. Let $\pi : \mathcal{X} \to [\underline{p}, \overline{p}]$ be a pricing policy that maps any continuous context vector $x \in \mathcal{X}$ to a price decision $p \in [\underline{p}, \overline{p}]$. We shall restrict ourselves to deterministic pricing policies only in this paper, because such policies are optimal when the demand model in Equation (1) is completely known and specified. The following definition specifies the classes of fair contextual pricing policies.

Definition 1 (Utility Fairness). A contextual pricing policy $\pi : \mathcal{X} \to [\underline{p}, \overline{p}]$ satisfies δ_0-utility fairness (δ_0-UF) for a small positive parameter $\delta_0 > 0$, if for any $x, x' \in \mathcal{X}$ it holds that $|\pi(x) - \pi(x')| \le \delta_0 |(x - x')^\top \theta_0|$.

Intuitively, Definition 1 requires the pricing policy π to offer similar prices to customers whose baseline utility values $x^\top\theta_0$ (excluding pricing effects) are similar, even though their context vectors or features x, x' could be very different. A pricing policy satisfying such fairness constraints will not discriminate against specific context components or features of a customer. Furthermore, it will explicitly set similar prices for customers with similar baseline utility values.

Note that, the personalized pricing practice *without* fairness constraints *does* charge higher prices for higher utility individuals and might be unfair. In this work, we adopt the δ_0-utility fairness (δ_0-UF) framework to *limit* the algorithm's price discrimination capabilities to foster fairness. In other words, a δ_0-UF pricing policy could still charge customer groups with higher utilities a higher price, but this ability is limited to a level of δ_0 rather than unlimited without fairness constraints. This helps the pricing algorithm to balance fairness constraints (in terms of the value of δ_0) while still being able to make personalized pricing decisions leading to improved profits compared with nonpersonal pricing that is completely fair but has low profit potentials.

4. Optimal Contextual Pricing Policies with Fairness Constraints

Given the definition of fairness constraints in the previous section, the optimal pricing policy that maximizes the expected revenue can be found by solving the following infinite-dimensional optimization problem:

$$\begin{aligned} \pi^* = \arg \max_{\pi:\mathcal{X}\to[\underline{p},\overline{p}]} & \mathbb{E}_{x\sim P_{\mathcal{X}}}[r_{x^\top\theta_0}(\pi(x))] \\ \text{s.t. } & |\pi(x) - \pi(x')| \le \delta_0 |(x - x')^\top \theta_0|. \quad \forall x, x' \in \mathcal{X}. \end{aligned} \tag{3}$$

Noting that both the objective and the right-hand side of the fairness constraints only involve the baseline utility value $u = x^\top\theta_0$, the optimal policy π^* can be reformulated into $\pi^*(x) = \varpi^*(x^\top\theta_0)$ such that

$$\begin{aligned} \varpi^* = \arg \max_{\varpi:[-B,B]\to[\underline{p},\overline{p}]} & \int_{-B}^{B} r_u(\varpi(u)) \mathrm{d}P_{\mathcal{U}}(u) \\ \text{s.t. } & |\varpi(u) - \varpi(u')| \le \delta_0 |u - u'|, \quad \forall u, u' \in [-B, B], \end{aligned} \tag{4}$$

where $P_{\mathcal{U}}$ is the marginal distribution of the baseline utility $u = x^\top\theta_0$, whose support is contained in $[-B, B]$ thanks to Assumption 1. Note that Equation (4) remains an infinite-dimensional optimization problem in search of an optimal function, but the fact that ϖ^* is a univariate function enables us to find approximate computational procedures and to gain deeper insights into the structures of the optimal policy.

4.1. Discretization and an Approximation Algorithm for Finding the Optimal Policy

We first present a general-purpose approximation algorithm based on discretization that approximately solves the infinite-dimensional problem in Equation (4). Let $\epsilon > 0$ be a small positive parameter for approximation errors and $\{u_j\}_{j=1}^{M_u}$, $\{p_j\}_{j=1}^{M_p}$ be discretized base utility values and prices, where $M_u = \lceil 2B/\epsilon \rceil$, $M_p = \lceil (\overline{p} - \underline{p})/(\delta_0\epsilon) \rceil$, and $u_{j+1} = u_j + \epsilon$, $p_{j+1} = p_j + \delta_0\epsilon$ for all j, such that $[-B, B] = \cup_{j=1}^{M_u} [u_j - \epsilon/2, u_j + \epsilon/2]$ and $[\underline{p}, \overline{p}] = \cup_{j=1}^{M_p} [p_j - \delta_0\epsilon/2, p_j + \delta_0\epsilon/2]$. Note that the grids $\{u_j\}, \{p_j\}$ are both depending on the discretization parameter ϵ: We are omitting ϵ subscripts to simplify notations but the fact that they are ϵ-dependent must be kept in mind. Consider the following finite-dimensional, discrete

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optimization problem:

$$j_1^*, \ldots, j_{M_u}^* = \arg \max_{j_1, \ldots, j_{M_u} \in [M_p]} \sum_{k=1}^{M_u} \gamma_k r_{u_k}(p_{j_k}) \quad \text{s.t. } |j_{k+1} - j_k| \le 1, \quad k = 1, 2, \ldots, M_u - 1, \tag{5}$$

where $\gamma_k := \Pr[x^\top \theta_0 \in [u_k - \epsilon/2, u_k + \epsilon/2]] = \int_{u_k-\epsilon/2}^{u_k+\epsilon/2} \mathrm{d}P_{\mathcal{U}}(u)$ is the probability of the baseline utility falling into $[u_k - \epsilon/2, u_k + \epsilon/2]$. With solutions $j_1^*, \ldots, j_{M_u}^* \in [M_p]$, an approximate contextual pricing policy could be constructed as $\pi_\epsilon^*(x) = \varpi_\epsilon^*(x^\top \theta_0)$ with

$$\varpi_\epsilon^*(u) = p_{j_{k(u)}^*} + \delta_0 \iota_{k(u)}(u - u_{k(x)}; \{j_k^*\}) \quad \text{where } k(u) = \arg \min_{k \in [M_u]} |u_k - u|, \tag{6}$$

where ties are broken arbitrarily in the argmax operator above, and the $\iota_k(\cdot) \in \{\pm 1\}$ functions are defined in the following ways:

$$\iota_k(w; \{j_k\}) := \begin{cases} +w, & \text{if } w > 0 \text{ and } j_{k+1} = j_k + 1, \\ & \quad \text{or } w < 0 \text{ and } j_{k-1} = j_k - 1; \\ -w, & \text{if } w > 0 \text{ and } j_{k+1} = j_k - 1, \\ & \quad \text{or } w < 0 \text{ and } j_{k-1} = j_k + 1; \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

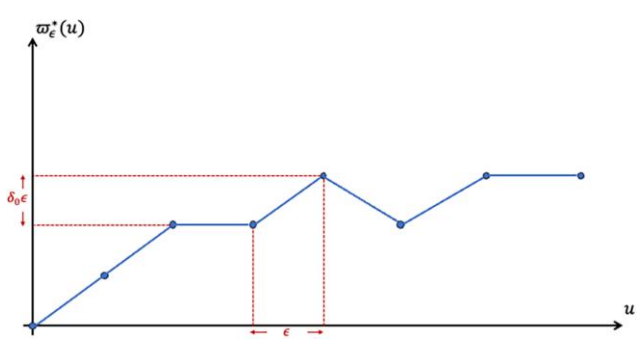
Intuitively, the constructed $\varpi_\epsilon^*(\cdot)$ is a piecewise linear interpolation of $\{u_k, p_{j_k^*}\}_k$, which must be δ_0-Lipschitz continuous thanks to the feasibility of the discrete price solutions. Figure 1 gives a graphical illustration of the constructed ϖ_ϵ^* and π_ϵ^* policy.

The following proposition shows that π_ϵ^* is fair and near optimal.

Lemma 1. *The policy π_ϵ^* satisfies δ_0-utility fairness constraint. Furthermore, it holds that*

$$\mathbb{E}_{x \sim P_{\mathcal{X}}}[r_{x^\top \theta_0}(\pi_\epsilon^*(x))] \ge \mathbb{E}_{x \sim P_{\mathcal{X}}}[r_{x^\top \theta_0}(\pi^*(x))] - 4L_f \delta_0 \epsilon.$$

Figure 1. (Color online) Construction of $\varpi_\epsilon^*(\cdot)$ from $\{j_k^*\}$



Notes. Constructed $\varpi_\epsilon^*(\cdot)$ corresponding to the sequence $j_1^* = 1$, $j_2^* = 2$, $j_3^* = 3$, $j_4^* = 3$, $j_5^* 4$, $j_6^* = 3$, $j_7^* = 4$, $j_8^* = 4$. Gaps between two consecutive points are ϵ on the x axis and $\delta_0\epsilon$ on the y axis. $\varpi_\epsilon^*(\cdot)$ is then a piecewise linear interpolation of all discrete utility-price points, which is δ_0-Lipschitz continuous thanks to the constraints on $\{j_k^*\}$. Finally, the implied optimal pricing policy is $\pi_\epsilon^*(x) = \varpi_\epsilon^*(x^\top \theta_0)$. Plot for illustration purposes only, and the plotted $\{j_k^*\}$ or $\varpi_\epsilon^*(\cdot)$ may not correspond to the optimal solution of an actual problem instance.

Lemma 1 shows that, if we can efficiently compute the optimal solution to Equation (6) for *every* fixed $\epsilon > 0$, and that the computational complexity is polynomial on both the problem sizer and $1/\epsilon$, then by taking $\epsilon \to 0^+$, we obtain a (fully) polynomial-time approximation scheme (FPTAS) of the infinite-dimensional optimization problem in Equation (4). Below we describe how such an efficient computation algorithm could be designed by using a dynamic programming approach.

In particular, for $k \in [M_u]$ and $j \in [M_p]$, let $V(k, j)$ be the maximized expected revenue up to u_k such that the last price is $j_k = j$. The Bellman's equation can then be written as

$$V(k+1, j) = \gamma_{k+1} r_{u_k}(p_j) + \max_{|j'-j| \le 1} V(k, j'), \tag{8}$$

with the initial condition that $V(1, j) = \gamma_1 r_{u_1}(p_j)$ for every j. Solving the dynamic program requires $O(\delta_0^{-1}\epsilon^{-2})$ space and time complexity, which achieves an $O(\epsilon)$-approximation thanks to Lemma 1.

To conclude this section, we mention some additional properties of the optimal δ_0-UF policy π^*. First, we show that if π^* is unique, then any δ_0-UF policy sequence that converges to π^* in expected revenue also converges to π^* uniformly at each utility value. Furthermore, we show that the expected revenue of the optimal δ_0-UF policy is left-continuous in δ_0. To avoid interrupting the flow of the paper, we place both additional technical results in the Online Appendix, which are nevertheless important in our proof of more important structural results later.

The term δ_0 should typically be chosen as a small but nonzero positive constant, usually much smaller than the intrinsic smoothness (Lipschitz continuity) of the demand reaction functions, to ensure a reasonable tradeoff between fairness and profit. Concrete determination of values of δ_0 is a subjective matter. As guidance, taking δ_0 to be a certain percentage (e.g., 50%) of the Lipschitz continuity parameter of a hypothetical fairness-unconstrained optimal pricing policy might be a reasonable idea.

4.2. Structures of the Optimal Fair Contextual Pricing Policy

In addition to the discretization and approximation algorithm proposed in the previous section that is general purpose and leads to an algorithm that is practical in most scenarios, in this section, we present additional structures in the optimal fair contextual pricing policy that offer more insights into the optimal policy while laying the foundation of follow-up analysis of the cost of fairness and a pricing policy with *incomplete information* that is more useful in practice when the demand curve is unknown in advance.

To derive the structures of the optimal fair pricing policy, we impose some additional assumptions on

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the link function f. In particular, we assume the existence of a parameter

$$\sigma_u := \inf_{u\in[-\widetilde{B},\widetilde{B}]} \left\{ f'(u) - \frac{\alpha_0\overline{p}}{2}|f''(u)| \right\} > 0 \tag{9}$$

that is strictly positive. To understand the high-level intuition behind Equation (9) (or more specifically the assumption that $\sigma_u > 0$, because a σ_u such defined as the infimum always exists), observe that an economically reasonable link function must satisfy $f'(u) > 0$ for all u because an increased utility means higher purchase probability and therefore higher expected demand. Therefore, Equation (9) is essentially assuming that either the price elasticity α_0 or the nonlinearity of the demand curve f'' is small. Below, we use several examples of popular demand functions to further illustrate the σ_u parameter.

Example 1 (Linear Demand). When the demand function is linear, we have the identity link function $f(u) = u$. Because the demand curve is perfectly linear and $f'' \equiv 0$ in this case, $\sigma_u = 1$ regardless of the value of price elasticity α_0.

Example 2 (Logistic Demand). When the demand function is logistic, we have $f(u) = e^u/(1+e^u)$. Equation (9) holds for every $f^{-1}\left(\max\left\{0, \frac{1}{2} - \frac{1}{\alpha_0\overline{p}}\right\}\right) < u < f^{-1}\left(\min\left\{1, \frac{1}{2} + \frac{1}{\alpha_0\overline{p}}\right\}\right)$, where f^{-1} is the inverse function of f. Furthermore, if $\alpha_0 \le 2/\overline{p}$, then Equation (9) holds for every $u \in \mathbb{R}$.

Example 3 (Exponential Demand). When the demand function is exponential, we have $f(u) = 1 - e^{-u}$ for $u \ge 0$. Equation (9) holds for every $u \ge 0$ provided that $\alpha_0 < 2/\overline{p}$.

Another important motivation for considering Equation (9) lies in the observation that, if $\sigma_u > 0$, the optimal price (without subjecting to any fairness constraints) is a *monotonically increasing* function of the baseline utility $u = x^\top\theta_0$, which is economically intuitive and practically reasonable because we are expected to offer (even only slightly) higher prices to individuals who value the offered products or service more. This intuition is rigorously summarized in the following lemma.

Lemma 2. *Let $\sigma_u > 0$ be defined in* Equation (9) *and $M_r := \sup_u |r''(u)| < \infty$, whose existence is implied by Assumption 1. For every $u \in [-B, B]$, let $p^*(u) = \arg\max_p r_u(p)$ which belongs to $(-\underline{p}, \overline{p})$ thanks to Assumption 2. Then for every $u \le u'$, it holds that*

$$p^*(u) + \frac{\sigma_u}{M_r}(u' - u) \le p^*(u') \le p^*(u) + \sqrt{\frac{4L_f}{\sigma_r}(u' - u)}.$$

Lemma 2 can be proved by taking partial derivatives of r with respect to both u and p and then applying the mean-value theorem. Because of space constraints, the complete proof of Lemma 2 is given in the Online Appendix.

With $\sigma_u > 0$ defined in Equation (9), $M_r = \sup_u |r''(u)| < \infty$, as well as all assumptions listed in Section 3, our next main result establishes the structure of the optimal contextual pricing policy satisfying δ_0-utility fairness.

Theorem 1. *Suppose that $\sigma_u > 0$ and the optimal contextual pricing policy $\pi^*(x) = \varpi^*(x^\top\theta_0)$ satisfying δ_0-UF is unique. Then ϖ^* satisfies the following properties:*

1. *ϖ^* is monotonically nondecreasing; that is, for every $u \le u'$ it holds that $\varpi^*(u) \le \varpi^*(u')$;*
2. *ϖ^* is differentiable almost everywhere, and $\varpi^*(u) = p^*(u)$ if ϖ^* is differentiable at $u \in (-B, B)$ and $\partial_u\varpi^*(u) < \delta_0$.*

Furthermore, if $\delta_0 > 0$ satisfies $\delta_0 \le \sigma_u/M_r$ then ϖ^* takes the form of

$$\varpi^*(u) = \pi_0 + \delta_0 u, \tag{10}$$

where $\pi_0 = \varpi^*(0) \in [\underline{p}, \overline{p}]$ is the price set for the case of zero baseline utility value.

Before discussing the consequences and proof of Theorem 1, we make a remark when the demand model $f(u) = u$ is the purely linear model. In particular, the following remark shows that for purely linear models, Theorem 1 completely characterizes the behavior/structure of the optimal δ_0-UF contextual pricing policy for *every* value of $\delta_0 > 0$.

Remark 1. When the demand curve is linear, Theorem 1 completely characterizes the structure of the optimal δ_0-UF contextual pricing policy for *all* values of δ_0. To see this, note that for a demand model of $f(u - \alpha_0 p) = u - \alpha_0 p$, the optimal price (without fairness constraints) for a baseline utility value u is $p^*(u) = u/\alpha_0$, which automatically satisfies δ_0-UF for if $\delta_0 \ge 1/\alpha_0$. On the other hand, note that in this case, $\sigma_r = 1$ and $M_r = \alpha_0$, and Theorem 1 specifies the structure of the optimal δ_0-UF contextual pricing policy for all $\delta_0 < 1/\alpha_0$. Combining both cases, we obtain

$$\varpi^*(u) = \begin{cases} \varpi^*(0) + \delta_0 u, & \text{if } \delta_0 < 1/\alpha_0; \\ p^*(u), & \text{if } \delta_0 \ge 1/\alpha_0. \end{cases} \quad \square$$

We next explain the conclusions and consequence of Theorem 1. The first two properties establish two important structures of ϖ^*, and they apply to all values of $\delta_0 > 0$. Intuitively, ϖ^* always sets the offered price as a monotonically increasing function of the baseline utility value, much in the same way of the fairness unconstrained policy in this case because $p^*(u)$ is a monotonically increasing function of u, thanks to Lemma 2. Additionally, if the slope of ϖ^* is strictly less than δ_0 at a certain point,[2] then the offered price under ϖ^* must coincide with $p^*(u)$, the optimal price without fairness considerations. This shows that,

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roughly speaking, ϖ^* either increases at a rate/slope of δ_0, the maximum allowed under δ_0-utility fairness, or ϖ^* must take the form of *unconstrained* contextual pricing when the utility fairness constraint is "not binding" at a certain point.

Theorem 1 also shows that, when the utility fairness parameter δ_0 is not too large, the optimal contextual pricing policy $\pi^*(x) = \varpi^*(x^\top\theta_0)$ takes a simple form of offering prices linearly increasing with the baseline utility value, with the slope being set exactly at δ_0.

Next, we give a high-level sketch of the proof of Theorem 1. The complete proof is very technical and therefore deferred to the Online Appendix. The idea is to discretize the infinite-dimensional optimization problem of ϖ^*, so that there are a finite number of (discretized) utility values. The prices for each utility value, different from the formulation of Equation (4), can vary continuously in $[\underline{p}, \overline{p}]$. This leads to a finite-dimensional, linear constrained optimization problem, in which strong duality holds thanks to Slater's condition. We then use complementary slackness and induction to analyze the signs of Lagrangian multipliers of utility fairness constraints at optimality, which eventually leads to the conclusions given in Theorem 1.

4.3. Cost of Utility Fairness

In this section, we study the "cost" of utility fairness constraints on the expected revenue when an optimal contextual pricing policy (subject to corresponding fairness constraints) is implemented. More specifically, for every $\delta_0 > 0$, recall the definition that

$$R(\delta_0) := \max_{\varpi:[-B,B]\to[\underline{p},\overline{p}]} \mathbb{E}_{P_{\mathcal{U}}}[r_u(\varpi(u))] \quad \text{s.t. } |\varpi(u) - \varpi(u')| \le \delta_0|u - u'|, \quad \forall u, u' \in [-B, B],$$

which is the expected revenue of the optimal contextual pricing policy that satisfies δ_0-UF. Clearly, $R(\delta_0)$ is a monotonically nondecreasing function of δ_0 because the feasible region gets larger as δ_0 increases, and $R(+\infty)$ is the expected revenue of the optimal contextual pricing policy *without* any fairness constraints. We then define

$$\rho(\delta_0) := \frac{R(\delta_0)}{R(+\infty)} \in [0, 1]$$

as the ratio between the optimal revenue at UF level δ_0 versus the optimal revenue without fairness constraints. This quantity characterizes the tradeoff between revenue and fairness constraints and also sheds light on the cost of utility fairness: A $\rho(\delta_0)$ value close to one means that the δ_0-UF constraint is not as penalizing compared with the revenue ideally obtainable without any privacy constraints, whereas on the other hand, a $\rho(\delta_0)$ value close to zero indicates that such UF requirement at the δ_0 level is probably too restrictive, because even the optimal contextual pricing policy subject to such fairness constraints suffers from a big gap from the ideal revenue without fairness constraints.

Our next theorem gives a closed-form expression of $\rho(\delta_0)$ when the demand model is purely linear.

Theorem 2. *Let $f(x) = x$ be the identity link function, corresponding to a purely linear demand model, and $\alpha_0 > 0$ being the price elasticity parameter. Then $\rho(\delta_0) = 1$ for all $\delta_0 \ge 1/\alpha_0$, and*

$$\rho(\delta_0) = \alpha_0\delta_0(2 - \alpha_0\delta_0) + (1 - \alpha_0\delta_0)^2\frac{\mu_u^2}{\nu_u^2}, \quad \forall \delta_0 < \frac{1}{\alpha_0},$$

where $\mu_u = \mathbb{E}_{P_{\mathcal{U}}}[u]$ and $\nu_u^2 = \mathbb{E}_{P_{\mathcal{U}}}[u^2]$.

Remark 2. Let $\kappa_u := \mu_u^2/\nu_u^2$. Because $\nu_u^2 - \mu_u^2$ is the variance of u under $P_{\mathcal{U}}$ that is nonnegative, we have that $\kappa_u \in [0, 1]$. Using elementary algebra calculations, we have that $\rho'(\delta_0) = 2\alpha_0(1 - \kappa_u)(1 - \alpha_0\delta_0)$, which is positive for all $\delta_0 \in (0, 1/\alpha_0)$, compatible with the intuition and property that ρ is a monotonically increasing function of u. Furthermore, because $\rho''(\delta_0) = -2\alpha_0^2(1 - \kappa_u) < 0$ for all $\delta_0 < 1/\alpha_0$, the function ρ is *concave* in δ_0 on $[0, 1/\alpha_0]$.

In general, Theorem 2 and Remark 2 together paint the complete picture of the impact of δ_0-utility fairness constraints the expected revenue of the (optimal) pricing policy. As δ_0 increases, the "relative efficiency" $\rho(\delta_0)$ gradually increases to one in a quadratic, concave way until δ_0 hits the threshold of $1/\alpha_0$ at which $\rho(\delta_0)$ saturates at $\rho(\delta_0) = 1$ for even larger δ_0 values (i.e., weaker fairness constraints). Graphical illustrations of Theorem 2 are given in the next section.

Theorem 2 only applies to the purely linear model $f(u) = u$. For more general nonlinear demand models, the complexity of the expected revenue and optimal pricing strategies makes analytical results such as Theorem 2 difficult to obtain. Nevertheless, we could still use numerical and computational studies to get some insights in this general case, which is covered in the next section.

4.4. Computational Results

We complement the results in the previous section with some computational results and graphical illustrations of the cost (in terms of reduced expected revenue) of utility fairness constraints.

In Figure 2, two plots of the relative revenue performance (of the best pricing policy subject to δ_0-UF) is plotted as a function of the fairness parameter δ_0, for the purely linear model $f(u) = u$. These plots are produced by instantiating the formula for $\rho(\cdot)$ derived in Theorem 2. We observe that under various settings and scenarios, $\rho(\cdot)$ is a monotonically increasing, concave curve. Furthermore, when the underlying distribution of context $P_{\mathcal{X}}$ is fixed, the left panel of Figure 2 shows that larger price elasticity α_0 leads to smaller "transition

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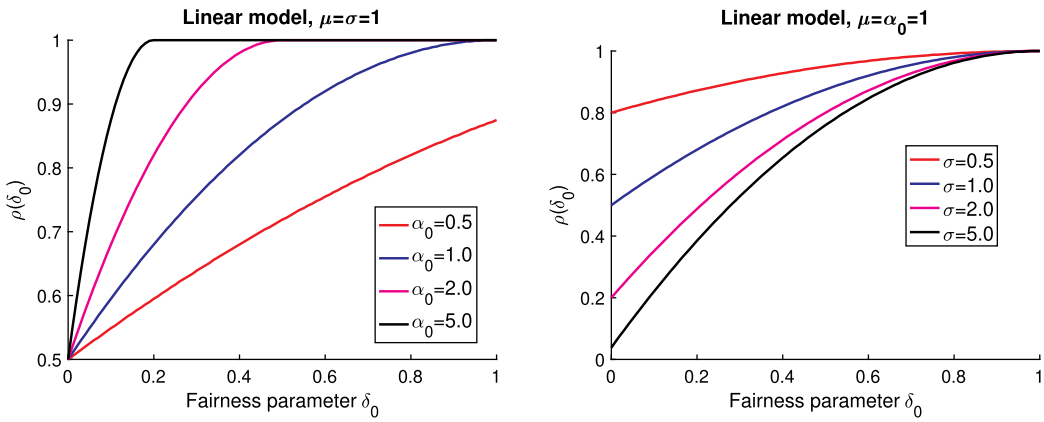
Figure 2. (Color online) Values of $\rho(\delta_0)$ for Purely Linear Models $f(u) = u$



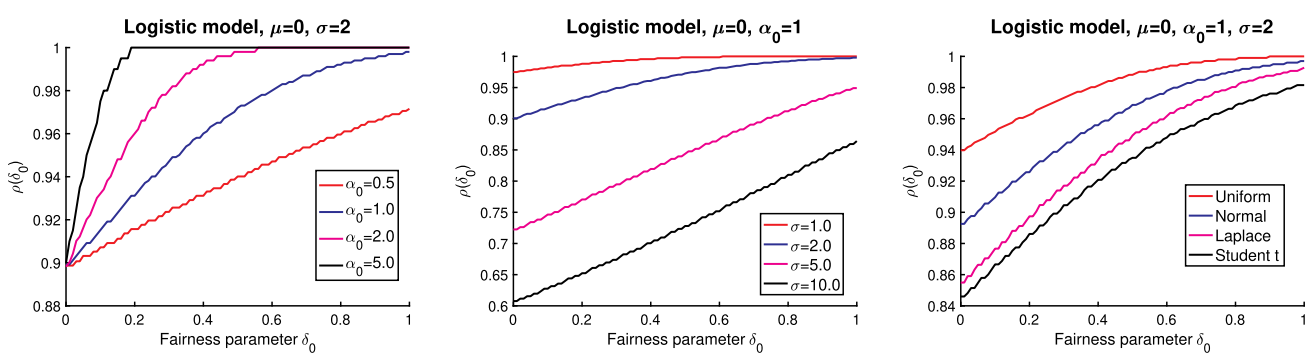
Notes. $\mu = \mathbb{E}_{P_X}[u]$ and $\sigma = \sqrt{\mathbb{E}_{P_X}[(u-\mu)^2]}$. By Theorem 2, for purely linear models only the mean and standard deviation of the baseline utility values are important. Left panel fixes $\mu = \sigma = 1$ and plots $\rho(\delta_0)$ as a function of δ_0 for different values of α_0, the price elasticity parameter. Right panel fixes $\mu = \alpha_0 = 1$ and plots the $\rho(\cdot)$ curve for different values of the standard deviation (of the baseline utility), σ.

points" of the fairness parameter δ_0 such that the revenue of a δ_0-UF policy converges to that of a nonfair one at smaller levels of δ_0, but steeper decays of the revenue when stricter fairness constraints are imposed (δ_0 below the transition points). The right panel of Figure 2 demonstrates the effects of different standard deviations of the underlying utility distribution, when fixing the mean of the utility distribution and the price elasticity parameter α_0 unchanged. In this case, when δ_0 increases $R(\delta_0)$ converges to the same point for all scenarios, but the decay of expected revenue is faster for distributions with larger standard deviation.

In Figure 3, three plots of the relative revenue performance (of the best pricing policy subject to δ_0-UF) is plotted as a function of the fairness parameter δ_0, for the logistic demand model $f(u) = e^u/(1+e^u)$. These plots are produced by running the FPTAS dynamic programming developed in Section 4.1, with 100 discretization points for utility values and 10,000 discretization points for prices. Note that, because of such discretization and approximation, calculations of $\rho(\delta_0)$ are approximate in Figure 3, and therefore small perturbations of the curves are spotted and expected.

The left panel of Figure 3 shows that the $\rho(\cdot)$ curves behave in a similar manner as the purely linear case, when one changes the price elasticity parameter α_0: A higher α_0 leads to lower "transition point" but faster decay of revenue below it. The middle panel of Figure 3 compares the $\rho(\cdot)$ curves with different standard deviation of baseline utility distributions while fixing other problem parameters. Compared with the right panel of Figure 2 for the purely linear case, several differences exist. First, these curves no longer converge to the same point as δ_0 increases, and the revenues of more variable distributions (larger standard deviation σ) are considerably lower. Additionally, unlike the purely

Figure 3. (Color online) Values of $\rho(\delta_0)$ for Logistic Demand Models $f(u) = e^u/(1+e^u)$



Notes. $\mu = \mathbb{E}_{P_X}[u]$ and $\sigma = \sqrt{\mathbb{E}_{P_X}[(u-\mu)^2]}$. Curves calculated using the FPTAS developed in Section 4.1, with further details in the main text. Because approximate calculations are used, the curves have small perturbations in all plots. Left panel plots $R(\delta_0)$ for different price elasticity parameters α_0, and middle panel plots $R(\delta_0)$ for different standard deviations of the underlying distribution. In both these plots, underlying distributions of baseline utility values are centered normal. Right panel plots $R(\delta_0)$ for fixed price elasticity, utility mean, and utility standard deviation, but different types of distributions (all centered): the uniform distribution, the normal distribution, the Laplace distribution, and the Student's t-distribution with degree-of-freedom being three.

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linear case when all curves are concave, for logistic models when σ is large, the $\rho_0(\cdot)$ curve becomes less concave and potentially nonconcave eventually, shedding lights on the complexity of the problem with nonlinear link functions.

Finally, the right panel of Figure 3 shows that, even if the underlying utility distribution shares the same mean and standard deviation, the actual shape of the distribution is important and different distributions (with the same mean and standard deviation) give rise to different $\rho(\cdot)$ curves. Furthermore, the plot suggests for distributions with heavier tails (Laplace, Student's t, etc.), the overall revenue curves are slightly lower than those associated with light-tailed distributions (uniform, Normal, etc.). This is an interesting finding and in stark contrast with the purely linear model, in which only the mean and the standard deviation of the underlying utility distribution would affect the revenue curve.

5. Fair Contextual Pricing with Incomplete Information

In this section, we present and analyze a dynamic contextual pricing algorithm (subject to approximate utility fairness constraints) when the demand model θ_0 and α_0 are *unknown* in advance and must be learned simultaneously with ongoing pricing decisions. Following the setup in the previous sections, we adopt the following generalized linear model

$$\mathbb{E}[y|x,p] = f(x^\top\theta_0 - \alpha_0 p), \tag{11}$$

where $\theta_0 \in \Theta \subseteq \mathbb{R}^d$, $\alpha_0 \in A \subseteq \mathbb{R}_+$, with Θ, A being known classes that contain the unknown model parameters.

To facilitate the design and analysis of bandit/learning algorithm under a utility fairness setting, we introduce the following additional technical assumption on the properties of the distribution $P_\mathcal{X}$ of the context vectors.

Assumption 4. *There exists a constant $\sigma_x > 0$ such that $\lambda_{\min}(\mathrm{Cov}(P_\mathcal{X})) \geq \sigma_x$, where $\mathrm{Cov}(P_\mathcal{X}) = \mathbb{E}_{P_\mathcal{X}}[xx^\top] - (\mathbb{E}_{P_\mathcal{X}}[x])(\mathbb{E}_{P_\mathcal{X}}[x])^\top$ is the covariance matrix of $x \sim P_\mathcal{X}$ and $\lambda_{\min}(\cdot)$ is the smallest eigenvalue of a positive semidefinite matrix.*

Intuitively, Assumption 4 states that the distribution of the context vectors $P_\mathcal{X}$ is *nondegenerate*, with its covariance matrix covering all directions in $\mathbb{R}^d$ and therefore equipped with a smallest eigenvalue bounded away from zero. We adopt Assumption 4 for our bandit algorithm and design for two reasons. First, it is a relatively intuitive and natural assumption and is satisfied whenever the context vectors are diverse instead of the rarer situation in which context vectors are concentrated on certain directions. Furthermore, without Assumption 4, it is not possible to estimate θ_0 in ℓ_2 norm (because some directions in $\mathbb{R}^d$ are not well covered by design), and therefore it will be very challenging to design δ_0-UF policies based on such deficient estimates because utility fairness is defined in a *uniform* manner, in which *all* pairs of customers' offered prices must be similar simultaneously.

The dynamic problem in which a decision maker adopts sequential personalized pricing decisions is formulated as below. At the beginning $t \in [T]$ of time period t, a total of $Q \geq 1$ customers arrive with different context vectors $x_{t1}, \ldots, x_{tQ} \overset{i.i.d.}{\sim} P_\mathcal{X}$. The decision maker then offers personalized prices $p_{t1} = \pi_t(x_{t1}), \ldots, p_{tQ} = \pi_t(x_{tQ})$ for the Q arriving customers from a policy π_t that must satisfy δ_0 utility fairness constraints. Afterward the purchasing activities for each customer, $y_{t1}, \ldots, y_{tQ}$, are realized and observed and recorded by the decision maker to improve their pricing policy for future time periods.

5.1. Lower Bound

We first establish a lower bound that demonstrates the fundamental difficulties of the fair contextual pricing problem with demand learning. The lower bound is information theoretical, meaning that it applies to *any* algorithm the implements fair pricing policies over T time periods with policy adjustments only made based on *past* observations and data. For simplicity, the lower bound instances are constructed for $Q = 1$ but the requirement on each π_t being δ_0-UF is maintained.

For $\delta_0 > 0$, $\theta_0 \in \Theta$, and $\alpha_0 \in A$, let $\mathcal{F}_{\theta_0,\alpha_0}(\delta_0)$ denote the class of all contextual pricing policies $\pi : \mathcal{X} \to [\underline{p}, \overline{p}]$ that satisfies δ_0-utility fairness (δ_0-UF) as defined in Definition 1. Our next theorem establishes an $\Omega(T^{2/3})$ lower bound on the optimal contextual pricing algorithm over T consecutive time periods, provided that such an algorithm uses δ_0-UF policies with high probability. Note that in the lower bound the constructed problem instances satisfy *all* assumptions made up to this point, including the covariance nondegeneracy assumption 4.

Theorem 3 ($T^{2/3}$ Regret Lower Bound). *Fix arbitrary $\delta_0 \in (0, 1/2]$ and let $\underline{C} > 0$ be a strictly positive constant potentially depending on δ_0. Let $f(u) \equiv u$ be the purely linear demand model. There exists model setup $\mathcal{X}$, $P_\mathcal{X}$, $[\underline{p}, \overline{p}]$, and function classes Θ, A satisfying Assumptions 1–4 such that, for any contextual pricing with demand learning algorithm over T time periods implementing pricing policies $\pi_1(\cdot), \ldots, \pi_T(\cdot)$, if such an algorithm satisfies*

$$\Pr[\pi_1, \ldots, \pi_T \in \mathcal{F}_{\theta_0,\alpha_0}(\delta_0)] \geq 0.95, \quad \forall \theta_0, \alpha_0 \in \Theta \times A,$$

then

$$\sup_{\theta_0,\alpha_0 \in \Theta \times A} \mathbb{E}_{\theta_0,\alpha_0}\left[\sum_{t=1}^{T} \max_{\pi \in \mathcal{F}_{\theta_0,\alpha_0}(\delta_0)} R_{\theta_0,\alpha_0}(\pi) - R_{\theta_0,\alpha_0}(\pi_t)\right] \geq \underline{C} \times T^{2/3},$$

where $R_{\theta,\alpha}(\pi) = \mathbb{E}[\pi(x)(x^\top\theta - \alpha\pi(x)) | x \sim P_\mathcal{X}]$ is the expected revenue of policy π.

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The $\Omega(T^{2/3})$ regret lower bound established in Theorem 3 is quite surprising, because virtually all contextual bandit or parametric bandit pricing problems admit regret upper bounds of $\widetilde{O}(\sqrt{T})$ or $O(\log T)$ (Rusmevichientong and Tsitsiklis 2010, Li and Huh 2011). To understand why the fair contextual pricing problem with demand learning admits an $\Omega(T^{2/3})$ regret lower bound, it is instructive to understand the structure of the optimal δ_0-UF policy (for δ_0 being not too large) specified in Theorem 1: The optimal price of an incoming user with context x is a linear function of x, which means that there is *perfect colinearity* between the context vectors and offered prices if the pricing strategy is optimal. This creates a dilemma: If near-optimal pricing policy is implemented, then the high colinearity between x and p means that we cannot estimate θ_0 and α_0 both accurately at the same time; on the other hand, if prices are given to circumvent such colinearity, such a pricing strategy must be far from optimal because it violates the structure derived in Theorem 1. In the Online Appendix, we make the above argument formal by explicitly constructing adversarial problem instances and using Pinsker's inequality together with other information theory tools to rigorously prove the $\Omega(T^{2/3})$ regret lower bound in Theorem 3.

5.2. Algorithm Design

Algorithm 1 (Contextual Pricing with Utility Fairness Constraints and Demand Learning)

Input: likelihood function $\mathcal{L}$, number of customers arriving each time period Q, price domain $[\underline{p},\overline{p}]$, utility fairness parameter $\delta_0 > 0$, time horizon T, algorithm parameters $\kappa_1, \kappa_2, K > 0$.

For $T_0 = \lceil T^{2/3} \rceil$ time periods, offer $\underline{p}$ or $\overline{p}$ with half probabilities each and all Q customers arriving at a time period receive either $\underline{p}$ or $\overline{p}$; ▷ price experimentation

$\widetilde{\delta}_0 = \max\{0, \delta_0 - \kappa_1/\sqrt{T_0}\}$; ▷ cushion of fairness for estimation error

$\widehat{\theta}, \widehat{\alpha} \leftarrow \arg\max_{\theta,\gamma} \sum_{t=1}^{T_0} \sum_{\ell=1}^{Q} \ln \mathcal{L}(y_{t\ell} | x_{t\ell}^\top \theta - 0.5\gamma p_t)$; ▷ maximum likelihood estimation

$\underline{\pi} \leftarrow \underline{p} - \widetilde{\delta}_0 \max_{t,\ell \le T_0, Q} x_{t\ell}^\top \widehat{\theta}$, $\overline{\pi} \leftarrow \overline{p} - \widetilde{\delta}_0 \min_{t,\ell \le T_0, Q} x_{t\ell}^\top \widehat{\theta}$;

Discretize $[\underline{\pi}, \overline{\pi}]$ into K evenly spaced prices $\{\pi_k\}_{k=1}^K$ and $r_k \leftarrow n_k \leftarrow 0$; ▷ initialization of MAB

for $t = T_0 + 1, \ldots, T$ **do**

  $k_t \leftarrow \arg\max_{k \in [K]} r_k/n_k + \kappa_2/\sqrt{n_k}$†; ▷ UCB construction

  Implement policy $\pi_t(x) = \mathrm{Trim}_{[\underline{p},\overline{p}]}(\pi_{k_t} + \widetilde{\delta}_0 x^\top \widehat{\theta})$ *;

  $r_{k_t} \leftarrow r_{k_t} + \sum_{\ell=1}^{Q} y_{t\ell} \pi_t(x_{t\ell})$, $n_{k_t} \leftarrow n_{k_t} + Q$; ▷ update revenue counts

end for

* $\mathrm{Trim}[a,b](u)$ takes the value of if $u < a$, b if $u > b$, and u if $a \le u \le b$;

† If for some k, $n_k = 0$, select k_t as the smallest such k.

Algorithm 1 gives a pseudo-code description of our proposed contextual pricing algorithm with demand learning, subject to utility fairness constraints. At a higher level, Algorithm 1 can be divided into two phases. The first phase consists of $T_0 = \lceil T^{2/3} \rceil$ time periods, during which price experimentation is carried out by offering nonpersonalized fixed price to all incoming customers, at one lower price p_L and one higher price p_U. In the algorithm, the two prices are set as $p_L = \underline{p}$ and $p_U = \overline{p}$, but they can be changed to any pair of prices in $[\underline{p}, \overline{p}]$ that are not too close to each other. After the price experimentation phase ends and observations are collected, a *maximum likelihood estimation* (MLE) procedure is carried out to obtain estimates $\widehat{\theta}, \widehat{\alpha}$ of the unknown model parameters θ_0, α_0. The MLE formulation involves a likelihood function $\mathcal{L}$ as an input to the algorithm, which should be designed for different link functions f. More details and examples of such a likelihood function will be given later.

The second phase of Algorithm 1 is an application of the upper confidence bound (UCB) algorithm for multiarmed bandit (MAB), with arms being K discretized "starting prices" in $[\underline{p}, \overline{p}]$. Each starting price π_k, $k = 1, 2, \ldots, K$ completely characterizes a contextual pricing policy $\pi(x) = \pi_k + \widetilde{\delta}_0 x^\top \widehat{\theta}$, which will be proved to satisfy δ_0-utility fairness with high probability. The number of discretized prices K will scale as a polynomial function of the time horizon T, which is specified in Theorem 4. Although such an approach does *not* usually lead to an $\widetilde{O}(\sqrt{T})$ regret upper bound, we note that because our regret target is $\widetilde{O}(T^{2/3})$ (thanks to Theorem 3), it is sufficient for our purpose.

Algorithm 1 requires a "likelihood function" $\mathcal{L}$ as input, which is designed according to the link function f. In general, this likelihood function coincides with the common understanding of likelihood functions in statistics, as shown by several examples later. To be fully general, we consider any $\mathcal{L}$ function as a valid likelihood or risk function as long as it satisfies Assumption 5. To simplify the notations, we use $\beta_0 = (\theta_0, \alpha_0) \in \mathbb{R}^{d+1}$ as the extended model vector.

Assumption 5 (Likelihood Function). *$\ln \mathcal{L}$ is twice continuously differentiable with respect to $\beta = (\theta, \alpha)$ on $\mathbb{R}^{d+1}$. Furthermore, there exist constants $\rho_L > 0$ and $0 < \sigma_L \le M_L < \infty$ such that for every $p \in [\underline{p}, \overline{p}]$ and $z = (x, -p)$, the following hold:*

1. $\mathbb{E}_{z,\beta_0}[\nabla_\beta \ln \mathcal{L}(y|z,\beta_0)] = 0$;
2. *For every $y \in \{0,1\}$ and $\beta \in \mathbb{R}^{d+1}$, $\nabla^2_{\beta\beta} \ln \mathcal{L}(y|z|\beta) \preceq 0$;*
3. *For every $y \in \{0,1\}$ and $\beta \in \mathbb{R}^{d+1}$ such that $\|\beta - \beta_0\|_2 \le \rho_L$, $\|\nabla_\beta \ln \mathcal{L}(y|z,\beta_0)\|_2 \le M_L$ and $-\nabla^2_{\beta\beta} \ln \mathcal{L}(y|z,\beta_0) \succeq \sigma_L z z^\top$.*

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The first property of Assumption 5 states that the true underlying parameter β_0 is the stationary point of $\ln\mathcal{L}$ in expectation; the second property shows that $\ln\mathcal{L}$ is concave in β; and the third property shows that if β is in a neighborhood of the true parameter β_0, the gradient of $\mathcal{L}$ is bounded almost surely and furthermore $\ln\mathcal{L}$ is locally strongly concave. When $\mathcal{L}$ is the likelihood of a generative model, $\nabla_\beta \ln\mathcal{L}$ corresponds to the *score function* and $-\nabla^2_{\beta\beta}\ln\mathcal{L}$ corresponds to the *Fisher's information*, and all properties in Assumption 5 are standard in statistics for regular generalized linear models.

Example 4 (Linear Demand). For linear demand $f(u) = u$, the likelihood function can be chosen as $\mathcal{L}(y|z,\beta_0) = e^{-(y-z^\top\beta_0)^2/2}$, which satisfies Assumption 5 with parameters $\rho_L = B$, $M_L = \mathrm{diam}(\mathcal{X})$ and $\sigma_L = 1$, where $\mathrm{diam}(\mathcal{X}) = \sup_{x,x'\in\mathcal{X}} \|x - x'\|_2$ is the diameter of $\mathcal{X}$.

Example 5 (Logistic Demand). For logistic demand $f(u) = e^u/(1+e^u)$, the likelihood function can be chosen as $\mathcal{L}(y|z,\beta_0) = ye^{z^\top\beta_0}/(1+e^{z^\top\beta_0}) + (1-y)/(1+e^{z^\top\beta_0})$, which satisfies Assumption 5 with parameters $\rho_L > 0$, $M_L = \mathrm{diam}(\mathcal{X})$ and $\sigma_L = 0.5f(B + \mathrm{diam}(\mathcal{X})\rho_L)$.

Example 6 (Exponential Demand). For exponential demand $f(u) = 1 - e^{-u}$, the likelihood function can be chosen as $\mathcal{L}(y|z,\beta_0) = y(1-e^{-z^\top\beta_0}) + (1-y)e^{-z^\top\beta_0}$, which satisfies Assumption 5 with parameters $\rho_L > 0$, $M_L = (e^{-u_{\min}}/(1+e^{-u_{\min}}))\mathrm{diam}(\mathcal{X})$ and $\sigma_L = e^{-B-\mathrm{diam}(\mathcal{X})\rho_L}$, where $u_{\min} = \inf_z z^\top\beta_0 > 0$.

With the second property of the likelihood function $\mathcal{L}$ in Assumption 5, Line 4 is a convex minimization problem and could be solved by any standard convex optimization packages for low- and intermediate-dimensional problems.

5.3. Fairness and Regret Analysis

In this section, we present the main theorem analyzing the fairness and revenue performance of Algorithm 1.

Theorem 4. *Suppose Assumptions 1–5 hold, and $\delta_0 \le \sigma_u/M_r$ so that the optimal (full-information) δ_0-UF policy admits the linear form in* Equation (10) *as proved in Theorem 1. Suppose also that Algorithm 1 is run with parameters $\kappa_1 = \frac{8M_r\mathrm{diam}(\mathcal{X})\sqrt{\ln(dT)}}{\min\{\sigma_x, 0.25(\overline{p}-\underline{p})^2\}\sigma_r}$, $\kappa_2 = 4\sqrt{\ln T}$, and $K = \lceil T^{1/3}\rceil$, where $\mathrm{diam}(\mathcal{X}) = \sup_{x,x'\in\mathcal{X}} \|x - x'\|_2$ is the diameter of $\mathcal{X}$. Let $\pi_1,\ldots,\pi_T$ be the contextual pricing policies implemented during the T time periods. With probability $1 - \widetilde{O}(T^{-2})$, the following hold:*

1. *(Fairness). All $\pi_1,\ldots,\pi_T$ satisfy δ_0 -utility fairness as defined in* Definition 1 *with respect to the underlying model θ_0 and α_0;*
2. *(Regret). $\sum_{t=1}^T \mathbb{E}_{x\sim P_X}[r_x(\pi^*(x)) - r_x(\pi_t(x))] \le (6L_f\widetilde{B}\kappa_1 + 4L_f\kappa_2)QT^{2/3}$, where π^* is the optimal δ_0-UF policy and $r_x(p) = pf(x^\top\theta_0 - \alpha_0 p)$ is the expected revenue at $x \in \mathcal{X}$ with offered price p.*

Theorem 4 contains two results. The first result is the *fairness* guarantee, which shows that with high probability *all* contextual pricing policies $\pi_1,\ldots,\pi_T$ implemented by Algorithm 1 over the entire T time periods satisfy δ_0-utility fairness. The failure probability $\widetilde{O}(T^{-2})$ is particularly small, meaning that it is unlikely to seen even one failure event over all T time periods. Therefore, when implemented, the firm can be very confident that all pricing decisions made satisfy required fairness constraints.

The second result of Theorem 4 analyzes the *revenue* performance by upper bounding the cumulative regret between Algorithm 1 and a benchmark optimal policy with full information. In general, the cumulative regret of Algorithm 1 is asymptotically on the order of $\widetilde{O}(T^{2/3})$, which is sublinear in time horizon T demonstrating meaningful learning and at the same time matches the information-theoretical lower bound established in Theorem 3.

5.4. Computational Results

We complement our theoretical findings with some computational results. Algorithm 1 is implemented in Julia, with plots made by Matlab. The two algorithmic parameters κ_1, κ_2, K are simply selected as $\kappa_1 = \sqrt{\ln(dT)}$, $\kappa_2 = \sqrt{\ln T}$, and $K = \lceil T^{1/3}\rceil$ which are on the same asymptotic order of the theoretical values of these parameters. The linear and logistic demand functions are studied, with likelihood functions $\mathcal{L}$ selected as specified in examples in the previous section.

In Figure 4, *relative regret* (the difference between the cumulative revenue of the optimal δ_0-UF policy and the bandit algorithm, normalized by the cumulative revenue of the optimal δ_0-UF policy itself) is plotted under various experimental settings. It can be seen that, under all settings, the relative regret converges to zero as the number of time horizon T increases, corroborating the theoretical sublinear regret results established in the previous section.

We further produce a log-log plot between the relative regret and the time horizon in Figure 5 to investigate the asymptotic relationship between the relative regret and T when T is large. The observations lie roughly on a straight line in the log-log plot, suggesting that the cumulative regret grows as a sublinear, polynomial function of T. Furthermore, simple linear regression estimates a slope of -0.28 based on the observations produced in Figure 5, which implies an $O(T^{0.72})$ cumulative regret scaling and is close to the $\widetilde{O}(T^{2/3})$ theoretical scaling proved in Theorem 4.

Finally, to complement our numerical results on synthetic data, we test the performance of our proposed

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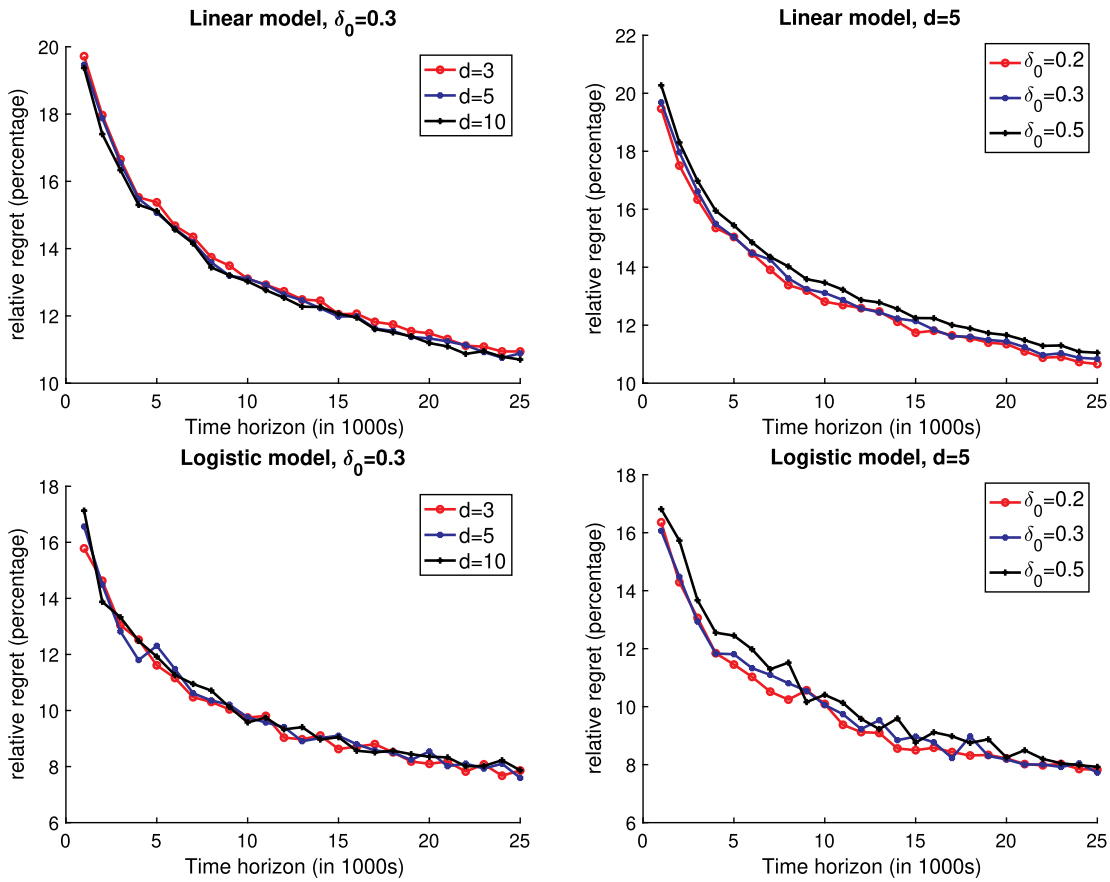
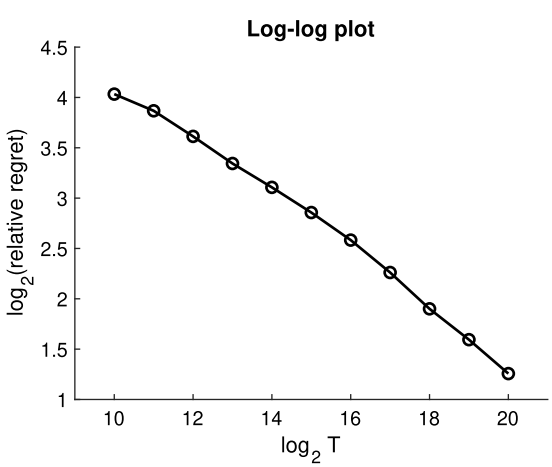
Figure 4. (Color online) Plots of Relative Regret vs. Time Horizon T



Notes. The first row is for purely linear demand and the second row is for logistic demand. The left panel is for different context dimensionality d, and the right panel is for different utility fairness parameters δ_0. The time horizon T ranges from 1,000 to 25,000. All experimental settings are repeated for 20 independent trials, and the mean relative regret is reported.

Figure 5. Log-Log Plot of Relative Regret vs. Time Horizon T



Notes. Logistic demand model, context dimension $d = 3$, utility fairness parameter $\delta_0 = 0.3$. The y axis is $\log_2$-relative regret, and the x axis is $\log_2 T$, where the time horizon T ranges from 2^{10} to 2^{20}. All experimental settings are repeated for 20 independent trials and the mean relative regret is reported.

algorithm on "semisynthetic" data based on a real-world online shopping data set.[3] We first estimate an 11-dimensional logistic regression model for demands using the data set and then run our algorithm using real contextual vectors from more than 50,000 users recorded in the data set, with hypothetical purchase behaviors randomly generated from the estimated 11-dimensional logistic demand models. Further experimental setup details are given in the Online Appendix.

In Figure 6, we report the relative regret of our proposed algorithm under various settings of time horizon T (ranging from 2,000 to 50,000 users) and utility fairness parameter δ_0 (ranging from 0.1 to 1.0). The result is consistent with our theoretical findings and numerical results on synthetic data, with the relative regret steadily decreasing as the number of time periods increases, and tradeoffs between strong utility fairness constraints (smaller δ_0 values) with profit performances (measured by relative regret) are also clear and expected in the results.

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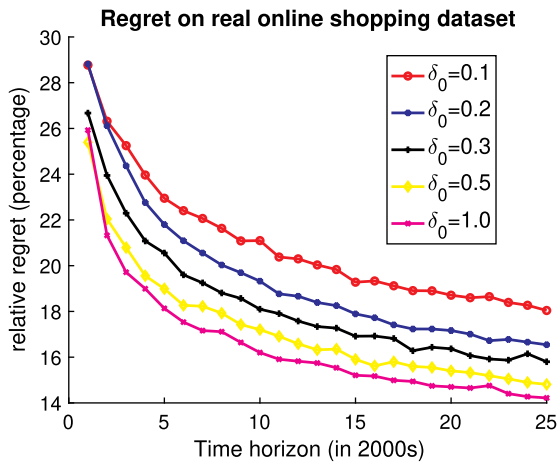
Figure 6. (Color online) Plots of Relative Regret on Real Online Shopping Data Set



Notes. Logistic demand models, context dimension $d = 12$, with the time horizon ranging from 2,000 to 50,000. All experimental settings are repeated for 20 independent trails, and the mean relative regret is reported.

6. Conclusions

In this paper, we developed a contextual bandit algorithm tailored for personalized pricing under the constraints of utility fairness and uncertain demand, achieving an optimal regret upper bound. Our investigation reveals the nuanced tradeoffs between fairness, utility, and revenue maximization, significantly contributing to the discourse in both dynamic pricing and fairness-ware machine learning fields. Several intriguing avenues for future research emerge from our study. Firstly, we aim to extend our exploration into dynamic pricing settings with more general demand functions, such as semiparametric or nonparametric demands. Additionally, examining the implications of various fairness constraints within the context of regulatory compliance presents an interesting challenge. Furthermore, conducting a longitudinal study to assess the long-term impacts of fairness-constrained pricing strategies on customer satisfaction, loyalty, and corporate reputation would provide valuable insights. Through these future research directions, we aim to deepen the understanding of fairness in dynamic pricing, fostering a more ethical and equitable landscape in algorithmic decision making.

From the perspective of fairness notions, it is important to note that the fairness notion implemented in the second part of this paper (dynamic contextual pricing with fairness constraints and demand learning) is a "local" notion that only ensures fairness at each time period when a particular pricing policy π_t is implemented; it, however, does not limit how policies could change across time periods, which might bring unexpected consequences from fairness viewpoints. A more general notion of fairness would incorporate such considerations and ask the *aggregated* pricing functions across multiple time periods to also satisfy fairness constraints, a *time-fairness* notion that was studied in existing works of Cohen et al. (2021) and Gupta and Kamble (2019). A technical challenge for achieving such results lies in effectively restricting the changes of policies implemented across time periods, which is potentially connected with limited switches research in bandit optimization literature (Simchi-Levi and Xu 2019).

Endnotes

[1] A function $f : \mathbb{R} \to \mathbb{R}$ is unimodal if there exists $u_0 \in \mathbb{R}$ such that $f(u_1) \le f(u_2)$ for any $u_1 \le u_2 \le u_0$, and $f(u_1) \ge f(u_2)$ for any $u_0 \le u_1 \le u_2$.

[2] The slope of ϖ^* can never exceeds δ_0 because of the definition of δ_0-utility fairness.

[3] See https://www.kaggle.com/datasets/jacksondivakarr/online-shopping-dataset/data.

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